# Macroeconomics I — MPE Insper

Practice list: The New Keynesian AS–AD Model (Lista 7)

*Solutions and grading rubric*

**Instructions.** Multiple choice: mark the single correct option (no penalty for wrong answers). *True/False*: every 2 wrong items cancel 1 right one (score = $\max\{0, \text{right} - \text{wrong}/2\}\times$ item value); blank items do not count. Discursive questions: show every step. Numbers are illustrative and chosen to give clean answers.
**Notation (Benigno 2015).** Lower-case letters are logs. AD: $y = \bar{y}_n + (g - \bar{g}) - \sigma[\, i - (\bar{p} - p) - \rho\,]$ (eq. 21, no consumption taxes). AS: $p - p^e = \kappa(y - y_n)$ (eqs. 17, 20). Natural output $y_n = \frac{1+\eta}{\sigma^{-1}+\eta}a + \frac{\sigma^{-1}}{\sigma^{-1}+\eta}g - \frac{1}{\sigma^{-1}+\eta}\mu$ (eq. 15); efficient output $y_e$ is the same without the $\mu$ term (eq. 19). Bars denote the long run.

## Block 1 — Foundations: AD, AS, natural and efficient output *(Benigno §§1–5 • 25 points)*

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**Question 1.** (10 points) Objective items.

**a)** (3 points) If the share of firms whose prices are fixed in advance rises, then:

A. $\kappa$ rises, the AS gets steeper, and a cut in $i$ moves output by more.

B. $\kappa$ falls, the AS gets flatter, and a cut in $i$ moves output by more.

C. $\kappa$ is unchanged, the AS shifts up, and a cut in $i$ moves prices by more.

D. $\kappa$ falls, the AS gets steeper, and a cut in $i$ moves prices by more.

**Answer:** B

**Step — slope.** Only flexible-price firms respond to marginal cost; with fewer of them the price level responds less to the output gap: $\kappa \downarrow$, a flatter AS, so shifts in AD move $y$ more and $p$ less.

**Ref:** Benigno (2015), §4 and §4.1, pp. 506–507.

**b)** (3 points) With $\sigma = \eta = 1$, a temporary rise of 0.04 in the (log) desired markup $\mu$ changes $(y_n,\, y_e)$ by:

A. $(-0.04,\, -0.04)$

B. $(0,\, -0.02)$

C. $(-0.02,\, 0)$

D. $(-0.02,\, -0.02)$

**Answer:** C

**Step — coefficients.** $\partial y_n/\partial\mu = -1/(\sigma^{-1} + \eta) = -\frac{1}{2}$, so $\Delta y_n = -0.02$; $y_e$ does not depend on $\mu$.

**Ref:** Benigno (2015), §4.4, eqs. (15) and (19), p. 508.

**c)** (2 points, T or F) In Benigno's model the AD slopes down because a higher price level raises money demand and therefore the nominal interest rate.

**Answer:** F

**Step — channel.** There is no LM curve: the central bank sets $i$. Given $\bar{p}$, a higher $p$ lowers expected inflation $\bar{p} - p$, raises the real rate $i - (\bar{p} - p)$, and households save more: an intertemporal channel through the Euler equation.

**Ref:** Benigno (2015), §3, eq. (21).

**d)** (2 points, T or F) After a permanent productivity increase, with $i$ unchanged, the economy ends up at $p = p^e$ and $y = y_n$: both curves shift and no policy action is needed.

**Answer:** T

**Step — §6.2.** $y_n$ and $\bar{y}_n$ rise by the same amount: AS moves right through $(p^e, y_n')$ and AD moves right by $\Delta\bar{y}_n$ (consumption smoothing). The natural rate is unchanged.

**Ref:** Benigno (2015), §6.2, p. 512.

**Question 2.** (15 points) Discursive. Copom's model. Take $\sigma = \eta = 1$, $\kappa = 0.5$, $\rho = 0.02$, $p^e = \bar{p} = 0$, $\bar{y}_n = 0$, $g = \bar{g}$, and initially $a = \mu = 0$, so $i = \rho$ gives $y = p = 0$.

**a)** (5 points) Solve AD and AS for equilibrium $y$ and $p$ as functions of $i$, $\bar{p}$, $p^e$, $y_n$ and $\bar{y}_n$. Which variables move AD and which move AS?

**Step — AD.** $y = X - \sigma p$ with $X \equiv \bar{y}_n + (g - \bar{g}) - \sigma(i - \rho) + \sigma\bar{p}$.

**Step — solve.** Substitute $p = p^e + \kappa(y - y_n)$: $y = \dfrac{X - \sigma p^e + \sigma\kappa\, y_n}{1 + \sigma\kappa}$, then $p = p^e + \kappa(y - y_n)$.

**Step — shifters.** AD: $i$, $g$, $\bar{g}$, $\bar{p}$, $\bar{y}_n$ (future $\bar{a}$, $\bar{\mu}$). AS: $p^e$ and $y_n$ (current $a$, $g$, $\mu$).

**Ref:** Benigno (2015), §4–5.

**Grading rubric.**

- Rewrites AD with $X$: 1 pt.
- Solves for $y$ and $p$: 2.5 pts.
- Correct list of shifters: 1.5 pts. Putting $\bar{p}$ on the AS: $-1$ pt.

**b)** (5 points) A temporary productivity gain: $a = 0.03$, $\bar{a} = 0$. Compute $y_n$, then $y$, $p$ and the gap with $i = \rho$. Which $i$ restores $p = p^e$ and $y = y_n$? Show that it violates the zero lower bound, compute the outcome at $i = 0$, and find the long-run price level $\bar{p}$ that the central bank must credibly promise to reach the target with $i = 0$.

**Step — natural.** $y_n = \frac{2}{2} \cdot 0.03 = 0.03$; $y_e = y_n$.

**Step — no action.** $X = 0$: $y = \frac{0.5 \cdot 0.03}{1.5} = 0.01$, $p = 0.5(0.01 - 0.03) = -0.01$; gap $= -0.02$.

**Step — required $i$.** Need $y = 0.03$, $p = 0$, so $X = 0.03$: $-(i - 0.02) = 0.03 \Rightarrow i = -1\% < 0$.

**Step — at the ZLB.** $X = 0.02$: $y = \frac{0.02 + 0.015}{1.5} = 0.0233$, $p = -0.0033$.

**Step — expectations.** With $i = 0$ we need $\sigma\bar{p} = 0.01$: $\bar{p} = 1\%$, giving $y = 0.03$, $p = 0$.

*Economic intuition.* The economy can produce more but demand does not rise with it today; the real rate must fall. When $i$ cannot, a higher promised future price level raises expected inflation and lowers the real rate, but only if the promise is believed.

**Ref:** Benigno (2015), §6.1 (p. 512) and §9.

**Grading rubric.**

- $y_n$ and outcome with $i = \rho$: 1.5 pts. Shifting AD in a *temporary* shock: max. 2 pts.
- Required $i$ and the ZLB: 1.5 pts.
- Outcome at $i = 0$ and the required $\bar{p}$: 2 pts.

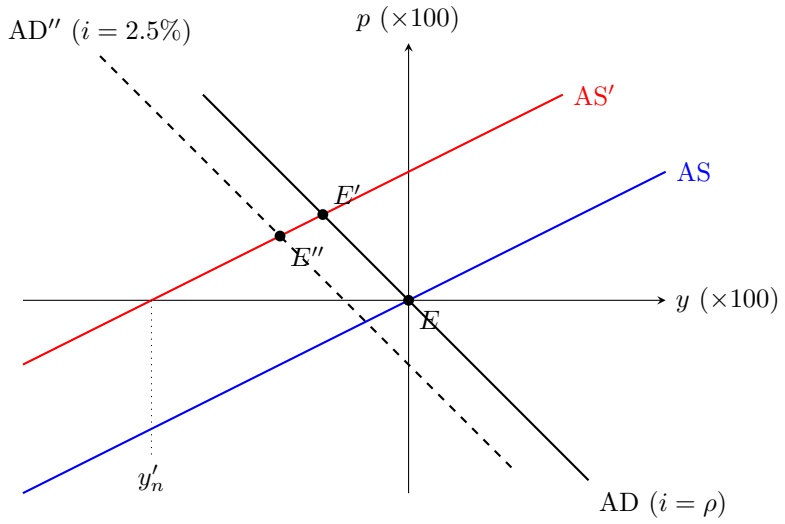
**c)** (5 points) A temporary markup shock (oil, a tax on firms): $\mu = 0.04$, $a = 0$. Compute $y_n$, $y_e$ and the outcome with $i = \rho$. Find $i$ under strict price stability. The bank minimizes $\mathcal{L} = (p - p^e)^2 + \lambda(y - y_e)^2$ subject to the AS, with $\lambda = 0.25$: derive the optimal $y$, $p$ and $i$. Draw the diagram.

**Step — levels.** $y_n = -0.02$, $y_e = 0$. With $i = \rho$: $y = \frac{0.5(-0.02)}{1.5} = -0.0067$, $p = 0.5(-0.0067 + 0.02) = +0.0067$: stagflation.

**Step — strict.** $p = 0 \Rightarrow y = y_n = -0.02$, so $X = -0.02$ and $i = \rho + 0.02 = 4\%$.

**Step — optimal.** Substitute the AS: $\min_y \kappa^2(y - y_n)^2 + \lambda y^2 \Rightarrow y = \dfrac{\kappa^2}{\kappa^2 + \lambda} y_n = \frac{1}{2}(-0.02) = -0.01$, $p = 0.5(-0.01 + 0.02) = 0.005$. From (a): $-0.015 = X - 0.01 \Rightarrow X = -0.005 \Rightarrow i = 2.5\%$.

**Step — trade-off.** Unlike (b), $y_n \neq y_e$: the bank cannot close the efficient gap and keep prices stable at once. It splits the shock according to $\lambda$.




$E = (y_e, p^e)$ is the origin; $E'$: no policy; $E''$: optimal policy; strict price stability would sit at $(y_n', p^e)$.

**Ref:** Benigno (2015), §7 (p. 513) and §11.

**Grading rubric.**

- $y_n$, $y_e$ and outcome at $i = \rho$: 1.5 pts.
- Strict price stability: 1 pt.
- Optimal $y$, $p$, $i$: 1.5 pts. Measuring the gap against $y_n$ in the loss: max. 2 pts in the item.
- Diagram with axes, AS shift through $(p^e, y_n')$, AD shift and points: 1 pt.

## Block 2 — Policy analysis

*(Benigno §§6–12 • 25 points)*

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**Question 3.** (10 points) Objective items.

**a)** (3 points) The Focus survey suddenly expects higher *future* productivity; current productivity is unchanged (Benigno §6.3). The policy that keeps $p = p^e$ and $y = y_n$ is to:

A. raise $i$, since AD shifts right while AS stays put.

B. cut $i$, since AS shifts right while AD stays put.

C. keep $i$, since $y_n$ today is unchanged by news.

D. promise a higher $\bar{p}$, since AD shifts left today.

**Answer:** A

**Step — §6.3.** Higher $\bar{a}$ raises $\bar{y}_n$: households want to consume more now (AD right). Current $y_n$ is unchanged (AS fixed), so output rises above both $y_n$ and $y_e$: the bank must *tighten*.

**Ref:** Benigno (2015), §6.3, p. 513.

**b)** (3 points) With $\sigma = \eta = 1$, $\kappa = 0.5$ and $i$ fixed, a temporary rise of 0.01 in $g$ raises output by:

A. 0.0100

B. 0.0050

C. 0.0150

D. 0.0083

**Answer:** D

**Step — both curves move.** AD: $X$ rises by 0.01. AS: $y_n$ rises by $\frac{\sigma^{-1}}{\sigma^{-1}+\eta} 0.01 = 0.005$. $y = \frac{0.01 + 0.5 \cdot 0.005}{1.5} = 0.0083$: multiplier $0.83 < 1$; the gap rises only 0.0033.

**Ref:** Benigno (2015), §8.

**c)** (2 points, T or F) At the zero lower bound, monetary policy has no instrument left to raise current output.
**Answer:** F

**Step — expectations.** A credible commitment to a higher long-run price level $\bar{p}$ raises expected inflation, lowers the real rate with $i = 0$ and shifts AD right.

**Ref:** Benigno (2015), §9.

**d)** (2 points, T or F) Productivity and government-spending shocks move $y_n$ and $y_e$ by the same amount, so they create no trade-off between stabilizing prices and the efficient gap.
**Answer:** T

**Step — eqs. (15), (19).** The coefficients on $a$ and $g$ are identical; only $\mu$ separates $y_n$ from $y_e$. Under flexible wages, "divine coincidence" holds for these shocks.

**Ref:** Benigno (2015), §4.4 and §11.

**Question 4.** (15 points) Discursive.

**a)** (5 points) Derive $y_n$ and $y_e$ from first-order conditions in log form. Use the labor supply condition $\eta l + \tilde{\sigma}^{-1} c = w - p$, the price rule of flexible firms $p = w - a + \mu$, the technology $y = a + l$ and the resource constraint $y = s_c c + g$, with $\sigma = \tilde{\sigma} s_c$. For $y_e$, replace the price rule by the planner's condition "marginal rate of substitution = marginal product of labor". Explain why a markup shock moves $y_n$ but not $y_e$.

**Step — substitute.** $c = (y - g)/s_c \Rightarrow \tilde{\sigma}^{-1} c = \sigma^{-1}(y - g)$; $l = y - a$.

**Step — natural.** Real wage $w - p = a - \mu$: $\eta(y - a) + \sigma^{-1}(y - g) = a - \mu$, so $y_n = \frac{(1+\eta)a + \sigma^{-1} g - \mu}{\sigma^{-1} + \eta}$.

**Step — efficient.** Planner: MRS = MPL (log MPL = $a$): $\eta(y - a) + \sigma^{-1}(y - g) = a$, the same without $\mu$.

**Step — why.** The markup is a wedge between the real wage and the marginal product of labor; the planner faces no such wedge. Monopoly power and distorting taxes lower natural output without changing what is efficient: $y_n - y_e = -\mu/(\sigma^{-1} + \eta)$.

**Ref:** Benigno (2015), §§3–4.4, eqs. (15) and (19).

**Grading rubric.**
- Substitutions: 1 pt.
- Derives $y_n$: 1.5 pts.
- Derives $y_e$: 1.5 pts.
- Wedge explanation: 1 pt. "The markup is inefficient" with no wedge argument: 0.5 pt.

**b)** (5 points) The Copom cuts the Selic by one percentage point. Compare the short- and long-run effects predicted by (i) the New Classical model, (ii) the original Keynesian (IS–LM) model and (iii) Benigno's New Keynesian model, along three dimensions: price adjustment and expectations; the policy instrument and how AD is built; and whether the model can rank outcomes by welfare.

**Step — New Classical.** Flexible prices, rational expectations, competitive markets. An anticipated cut is neutral; only surprises move output (Lucas). No welfare-relevant inefficiency to correct.

**Step — Keynesian.** Fixed or adaptive prices, AD from IS–LM with the money supply as the instrument. The cut raises output persistently; the model is not built from optimizing agents and cannot rank outcomes by household welfare.

**Step — New Keynesian.** Rational expectations, monopolistic competition, a fraction of prices preset. The instrument is $i$; AD comes from the Euler equation. Short run: AD shifts right, $y$ and $p$ rise along the AS. Long run: all prices adjust, $y = y_n$, money is neutral. Welfare is the household's utility, approximated by the quadratic loss in $(p - p^e)$ and $(y - y_e)$.

**Ref:** Benigno (2015), §1 and §12; Lista 7, Q1; Carlin & Soskice (2024) for the IS–LM contrast.

**Grading rubric.**
- Each framework with the three dimensions: 1.5 pts each (4.5 pts).

- Short- vs long-run distinction in the NK model: 0.5 pt.
- Building NK AD from LM / money demand: max. 3 pts in the item.

**c)** (5 points) A deleveraging shock (Eggertsson–Krugman): the debt limit of borrowers falls sharply. Explain why the economy can fall into a liquidity trap, why the fiscal multiplier becomes large, and why more price flexibility can make the recession worse (paradox of flexibility). Under what condition on long-run prices does the paradox disappear?

**Step — trap.** Borrowers must cut consumption; AD falls so much that the natural real rate turns negative. With $i \geq 0$ the bank cannot lower the real rate enough: output stays below $y_n$.

**Step — multiplier.** Ricardian equivalence fails: borrowers are constrained and spend extra income, and at $i = 0$ there is no crowding-out through higher rates. The multiplier can exceed one (Benigno reports 1.29 to 1.62), against the upper bound of one in §8.

**Step — flexibility.** Debt is nominal. If inflation between the short and the long run is fixed (as in Eggertsson–Krugman), a lower $p$ also lowers $\bar{p}$ and raises the real value of debt: borrowers cut consumption, and the AD can slope *up*. A steeper AS (more flexible prices) then means a larger fall in prices and a larger fall in output. For the same reason a downward AS shift (higher productivity, lower markups or taxes) becomes contractionary: the paradox of toil.

**Step — caveat.** If the long-run price level is anchored whatever happens to current prices, the AD slopes down again and both paradoxes disappear (Benigno, footnote 20).

*Economic intuition.* In a debt deflation, falling prices transfer resources to savers with a low propensity to spend, away from borrowers who must cut back.

**Ref:** Benigno (2015), §9 and §10, eq. (32), pp. 518–521.

**Grading rubric.**

- Liquidity trap mechanism with negative natural rate: 2 pts.
- Large multiplier with both reasons (constrained borrowers, no crowding-out): 1.5 pts.
- Paradox of flexibility through the real value of debt: 1.5 pts. Arguing through the real interest rate with a fixed $\bar{p}$: max. 0.5 pt on this line.